

NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA

THEORY EXAMINATION

Month and Year of Examination AUG 2026

Programme: B.Tech. (PIE)

Subject: Operations Research

Course No. PIPC-210

Semester: IV

Max. Marks: 50

Time allowed: 3 Hrs.

No. of pages used: 03

Notes: Attempt in total five questions. Do write the basic steps involved in solving a problem and demonstrate the computations involved.

- Q 1. A Company is manufacturing three products P1, P2 and P3. The minimum number of units of P1 and P3 that must be produced are **100 and 150** respectively. These products require two raw materials M1 and M2 which the firm can purchase up to a maximum of **500 and 400** units respectively. The profit earned by selling products P1, P2 and P3 is **Rs. 2, Rs. 5 and Rs. 4** respectively. The consumption of raw materials among the products is given below (*Use any kind of simplex method*)

Raw Material	Consumption of raw material per product		
	P1	P2	P3
M1	1/2	1	1
M2	2	1/2	1/5

Formulate the problem and find the PRODUCT MIX to *maximize* the net profit.

(10)

- Q 2. Tata Steel has three options with regarding to production:

(i) Produce commercially (ii) build pilot plant (iii) stop producing steel.

The management has estimated that their pilot plant, if built, has 0.75 chance of high yield and 0.25 chance of low yield. If the pilot plant does show a high yield, management assigns a probability of 0.7 that the commercial plant will also have a high yield. If pilot plant shows a low yield, there is only 0.15 chance that commercial plant will show high yield. Finally, management's best assessment of the yield on a commercial sized plant without building a pilot plant first has a 0.65 chance of high yield. A pilot plant will cost Rs. 3, 00, 000. The payoffs under high and low yield conditions are Rs. 1, 20, 00, 000 and Rs. - 12, 00, 000 respectively. Find the optimum decision for the company.

(10)

(Negative)

- Q 3. Products 1, 2, 3, 4, 5, 6 are to be processed on a machine. The set-up costs (in Rs. Per change) depend on the product presently on the machine and the product to be processed next, the data of which has been shown in table. Find the optimum sequence of products in order to minimize the total set-up cost using *branch and bound (travelling salesmen)* method.

(10)

	1	2	3	4	5	6
1	-	12	8	6	5	5
2	9	-	12	5	13	11
3	6	15	-	7	10	8
4	4	9	10	-	6	9
5	5	13	7	6	-	3
6	5	11	8	6	5	-

- Q 4. Find the optimum solution to the following transportation problem in which the cells contain the *Profit* in rupees. Obtain *ibfs* by using **least cost method** (3+7)

	W1	W2	W3	W4	W5	Supply
F1	32	65	96	26	85	18
F2	43	12	09	97	92	17
F3	09	40	25	55	40	19
F4	48	47	76	93	18	13
F5	49	82	18	87	93	15
Dem and	16	18	20	14	14	82(Total)

- Q 5. a) Discuss the interpretation of the following (**any two**): (2x2)
- (i) Basic and non-basic variables (ii) Basic solution (iii) Slack variables
- b) A factory manufactures three products A, B and C for which the data is given in the table below. The profit/unit is Rs. 32, 30 and 40 for these products (A, B and C) respectively.

	Product			Available Resources
	A	B	C	
Material needed (kg/unit)	5	4	3	2500 kg
Machine hours/unit	2	3	1	1275 hours
Labour hours/unit	3	2	4	2100 hours

The following **optimal table** is obtained after iterations-

Basis	X_1	X_2	X_3	S_1	S_2	S_3	Sol.
Z	3	0	0	0	4	9	24000
S_1	3/2	0	0	1	-1	-1/2	125
X_2	1/2	1	0	0	2/5	-1/10	300
X_3	1/2	0	1	0	-1/5	3/10	375

Find the change in the optimal solution and also establish the feasibility ranges if-

- (i) Materials availability improves by 125 units i.e. gets revised to 2625
 (ii) Machine hours available increase by 50, i.e. get revised to 1325
 (iii) Labour hours decrease by 75, i.e. get revised to 2025 (6)

- Q6. A company has one surplus truck in each of the cities A, B, C, D and E and one deficit truck in each of the cities 1,2,3,4,5 and 6. The distance between cities in KM is shown in the matrix below. Find the assignment of trucks from cities in surplus to cities in deficit so that the total distance covered by the vehicles is minimum. Also formulate the problem first. (2+8)

	1	2	3	4	5	6
A	12	10	15	22	18	8
B	10	18	24	15	16	12
C	11	10	3	8	5	9
D	6	14	10	13	12	12
E	8	12	11	6	13	10